

Project Summary: Supply & Demand Monitoring Platform

Overview

This project is a web-based platform that tracks product availability (supply) and user interest (demand) across different locations. It is NOT an e-commerce system — it does not handle buying or payments. Its purpose is to provide insights that help users and businesses make better decisions.

Core Features

- Product Search: Users can search for products (e.g., rice, fuel)
- Location-Based Results: Show where products are available
- Supply Data: Display quantity available per location
- Demand Data: Based on number of searches or user interactions
- Price Display: Show average price per location
- Delivery Estimation: Basic estimated delivery time
- Dashboard: Simple UI showing supply vs demand trends

Data Handling

- Supply Data: Input by vendors or manually added
- Demand Data: Based on number of searches per product per location
- Pricing: Use average of submitted prices and ignore extreme values

Price Control Logic

- Set minimum and maximum price range
- Calculate average or median price
- Flag unusual prices
- Optional: allow users to report incorrect prices

Technology Stack

- Frontend: React.js, HTML, CSS

- Backend: Node.js (Express.js)
- Database: Supabase (PostgreSQL)
- Tools: Git/GitHub

Basic System Flow

- User searches for a product
- Backend fetches data from database
- System returns locations, supply levels, demand levels, and average price
- Frontend displays results

Important Notes

- Focus on functionality, not perfection
- Data can be simulated
- UI should be simple and clean
- No payment system required

Goal

Build a working prototype that demonstrates supply vs demand tracking, location-based insights, and basic price fairness logic.

Future Improvements

- Real-time API integration
- Vendor authentication
- Advanced analytics (charts, predictions)
- Machine learning for demand forecasting